

Reality Manifest

Generated: 2026-07-18T18:14:43 Project: /private/tmp/ams-init

Metrics

- **total_python_files:** 34

Functional Blocks

F1: Cli [ACTIVE]

Source: auto-discovered from src/architecture_model/cli/

- **F1.1** Architecture Model Standard — universal machine [dormant] (src/architecture_model/init.py, 31 LOC)
- **F1.2** Package entry point: python [dormant] (src/architecture_model/main.py, 6 LOC)
- **F1.3** Init [dormant] (src/architecture_model/cli/init.py, 0 LOC)
- **F1.4** CLI for the Architecture Model Standard [active] (src/architecture_model/cli/main.py, 385 LOC)
 - main(argv: list[str] | None) -> int
- **F1.5** Configuration subpackage for the Architecture Model Standard [dormant] (src/architecture_model/config/init.py, 25 LOC)
- **F1.6** Configuration loader for the Architecture Model Standard [active] (src/architecture_model/config/loader.py, 558 LOC)
 - load_config(root: Path) -> ProjectConfig
 - discover_config(root: Path) -> tuple[ProjectConfig, DiscoveryReport]
 - get_config(root: Path) -> ProjectConfig
 - write_config(config: ProjectConfig, root: Path | None) -> Path
- **F1.7** Project configuration schema [active] (src/architecture_model/config/schema.py, 325 LOC)
- **F1.8** Init [dormant] (src/architecture_model/core/init.py, 0 LOC)
- **F1.9** Coverage analysis: compare architecture model against code reality (manifest) [active] (src/architecture_model/core/coverage.py, 268 LOC)
 - coverage_report(model: 'ArchitectureModel', manifest: dict) -> CoverageResult
- **F1.10** Complexity scoring and system identification for architecture decomposition [active] (src/architecture_model/core/decomposer.py, 659 LOC)
 - compute_complexity(comp: Component, model: ArchitectureModel) -> float
 - identify_systems(model: ArchitectureModel, manifest: dict) -> list[SystemCandidate]
 - auto_assign_f_blocks(model: ArchitectureModel, max_cluster_size: int) -> ArchitectureModel
 - decompose_model(model: ArchitectureModel, manifest: dict, output_dir: str) -> DecompositionResult
 - test_affinity_decompose(repo_path: Path) -> list[Subsystem]
- **F1.11** Differ: Compare two architecture model versions and produce a structured diff [active] (src/architecture_model/core/differ.py, 294 LOC)
 - diff_models(old_model: ArchitectureModel, new_model: ArchitectureModel) -> ModelDiff
- **F1.12** Merger: Merge manifest facts into an existing architecture model [active] (src/architecture_model/core/merger.py, 778 LOC)
 - merge_manifest(model: ArchitectureModel, manifest_path: str | Path, project_root: str | Path | None) -> ArchitectureModel
 - enrich_from_manifest(model: ArchitectureModel, manifest: dict) -> EnrichmentResult
 - compact_for_generation(model: ArchitectureModel) -> ArchitectureModel

- compose_enriched_model(project_root: Path) -> ArchitectureModel
- **F1.13** Parser [active] (src/architecture_model/core/parser.py, 592 LOC)
 - load_model(path: str | Path) -> ArchitectureModel
 - validate_model_data(data: dict[str, Any]) -> list[str]
 - dump_model(model: ArchitectureModel) -> dict[str, Any]
 - save_model(model: ArchitectureModel, path: str | Path) -> None
- **F1.14** Slicer: Extract subsets of the architecture model [active] (src/architecture_model/core/slicer.py, 496 LOC)
 - slice_by_fblock(model: ArchitectureModel, f_block: str, include_relationships: bool) -> ArchitectureModel
 - slice_by_layer(model: ArchitectureModel, layer_id: str) -> ArchitectureModel
 - slice_by_status(model: ArchitectureModel, status: Status) -> ArchitectureModel
 - slice_for_artifact(model: ArchitectureModel, artifact_name: str) -> ArchitectureModel
- **F1.15** Typed dataclasses for all Architecture Model entity types and relationships [active] (src/architecture_model/core/types.py, 665 LOC)
- **F1.16** Validator: Check architecture model invariants [active] (src/architecture_model/core/validator.py, 518 LOC)
 - validate_model(model: ArchitectureModel, strict: bool) -> ValidationResult
- **F1.17** Extract an ArchitectureModel directly from source code analysis [active] (src/architecture_model/extract/from_code.py, 646 LOC)
 - extract_from_code(project_root: str | Path, config: ProjectConfig | None, manifest: dict | None) -> ArchitectureModel
- **F1.18** Reality Manifest Generator — modular package [active] (src/architecture_model/manifest/init.py, 58 LOC)
- **F1.19** Functional block definitions and processing [active] (src/architecture_model/manifest/blocks.py, 215 LOC)
 - process_block(root: Path, block_id: str, block_def: dict, sub_block_configs: list | None) -> BlockManifest
- **F1.20** Body Hints [active] (src/architecture_model/manifest/body_hints.py, 270 LOC)
 - classify_function(source: str, func_name: str) -> BodyComplexity
 - extract_body_hint(source: str, func_name: str, class_name: str | None) -> str
 - extract_file_hints(filepath: Path, include_private: bool) -> list[FunctionSignature]
- **F1.21** Terminal summary display for the reality manifest [active] (src/architecture_model/manifest/display.py, 51 LOC)
 - print_summary(manifest: dict[str, Any]) -> None
- **F1.22** Main manifest generation and caching logic [active] (src/architecture_model/manifest/generator.py, 168 LOC)
 - generate_manifest(project_root: Path, config: Optional[Any]) -> Manifest
 - load_or_generate_manifest(project_root: Path, output_dir: Path | None) -> dict[str, Any]
- **F1.23** Interface derivation from import analysis between scanned modules [active] (src/architecture_model/manifest/interfaces.py, 89 LOC)
 - derive_interfaces(modules: list[ModuleInfo], root: Path) -> list[InterfaceEdge]
- **F1.24** Project metrics computation from filesystem scanning [active] (src/architecture_model/manifest/metrics.py, 85 LOC)
 - compute_metrics(root: Path, config: Optional[Any]) -> MetricsResult
- **F1.25** AST helpers and file scanning for the reality manifest [active] (src/architecture_model/manifest/scanner.py, 496 LOC)
 - scan_file(root: Path, filepath: Path) -> ModuleInfo
- **F1.26** Manifest slicing for artifact context injection [active] (src/architecture_model/manifest/slicers.py,

- 579 LOC)
 - get_manifest_slice(manifest: Manifest | dict[str, Any], artifact_name: str) -> str
- **F1.27** Test Contract Analyzer — extracts behavioral specifications from test files [active] (src/architecture_model/manifest/test_analyzer.py, 501 LOC)
 - analyze_test_file(test_file: Path) -> TestAnalysisResult
 - extract_constants_from_contracts(contracts: list[TestContract]) -> list[Constant]
- **F1.28** Typed dataclasses for manifest generation outputs [active] (src/architecture_model/manifest/types.py, 219 LOC)
- **F1.29** Domain profiles for cross [dormant] (src/architecture_model/profiles/init.py, 1 LOC)
- **F1.30** Init [dormant] (src/architecture_model/profiles/builtins/init.py, 0 LOC)
- **F1.31** Domain profile schema and loading logic [active] (src/architecture_model/profiles/schema.py, 85 LOC)
 - load_profile(name_or_path: str) -> DomainProfile
- **F1.32** Init [dormant] (src/architecture_model/spec/init.py, 0 LOC)
- **F1.33** Shared utilities for architecture [dormant] (src/architecture_model/utils/init.py, 1 LOC)
- **F1.34** Shared file discovery and exclusion logic [active] (src/architecture_model/utils/discovery.py, 72 LOC)
 - is_excluded_dir(path: Path) -> bool
 - collect_py_files(directory: Path, recursive: bool, exclude_init: bool) -> list[Path]
 - discover_source_files(project_root: Path) -> list[Path]
 - discover_test_files(project_root: Path) -> list[Path]

F2: Config [ACTIVE]

Source: Configuration subpackage for the Architecture Model Standard.

- **F2.1** Architecture Model Standard — universal machine [dormant] (src/architecture_model/init.py, 31 LOC)
- **F2.2** Package entry point: python [dormant] (src/architecture_model/main.py, 6 LOC)
- **F2.3** Init [dormant] (src/architecture_model/cli/init.py, 0 LOC)
- **F2.4** CLI for the Architecture Model Standard [active] (src/architecture_model/cli/main.py, 385 LOC)
 - main(argv: list[str] | None) -> int
- **F2.5** Configuration subpackage for the Architecture Model Standard [dormant] (src/architecture_model/config/init.py, 25 LOC)
- **F2.6** Configuration loader for the Architecture Model Standard [active] (src/architecture_model/config/loader.py, 558 LOC)
 - load_config(root: Path) -> ProjectConfig
 - discover_config(root: Path) -> tuple[ProjectConfig, DiscoveryReport]
 - get_config(root: Path) -> ProjectConfig
 - write_config(config: ProjectConfig, root: Path | None) -> Path
- **F2.7** Project configuration schema [active] (src/architecture_model/config/schema.py, 325 LOC)
- **F2.8** Init [dormant] (src/architecture_model/core/init.py, 0 LOC)
- **F2.9** Coverage analysis: compare architecture model against code reality (manifest) [active] (src/architecture_model/core/coverage.py, 268 LOC)
 - coverage_report(model: 'ArchitectureModel', manifest: dict) -> CoverageResult
- **F2.10** Complexity scoring and system identification for architecture decomposition [active] (src/architecture_model/core/decomposer.py, 659 LOC)
 - compute_complexity(comp: Component, model: ArchitectureModel) -> float
 - identify_systems(model: ArchitectureModel, manifest: dict) -> list[SystemCandidate]
 - auto_assign_f_blocks(model: ArchitectureModel, max_cluster_size: int) ->

- ArchitectureModel
 - decompose_model(model: ArchitectureModel, manifest: dict, output_dir: str) -> DecompositionResult
 - test_affinity_decompose(repo_path: Path) -> list[Subsystem]
- **F2.11** Differ: Compare two architecture model versions and produce a structured diff [active] (src/architecture_model/core/differ.py, 294 LOC)
 - diff_models(old_model: ArchitectureModel, new_model: ArchitectureModel) -> ModelDiff
- **F2.12** Merger: Merge manifest facts into an existing architecture model [active] (src/architecture_model/core/merger.py, 778 LOC)
 - merge_manifest(model: ArchitectureModel, manifest_path: str | Path, project_root: str | Path | None) -> ArchitectureModel
 - enrich_from_manifest(model: ArchitectureModel, manifest: dict) -> EnrichmentResult
 - compact_for_generation(model: ArchitectureModel) -> ArchitectureModel
 - compose_enriched_model(project_root: Path) -> ArchitectureModel
- **F2.13** Parser [active] (src/architecture_model/core/parser.py, 592 LOC)
 - load_model(path: str | Path) -> ArchitectureModel
 - validate_model_data(data: dict[str, Any]) -> list[str]
 - dump_model(model: ArchitectureModel) -> dict[str, Any]
 - save_model(model: ArchitectureModel, path: str | Path) -> None
- **F2.14** Slicer: Extract subsets of the architecture model [active] (src/architecture_model/core/slicer.py, 496 LOC)
 - slice_by_fblock(model: ArchitectureModel, f_block: str, include_relationships: bool) -> ArchitectureModel
 - slice_by_layer(model: ArchitectureModel, layer_id: str) -> ArchitectureModel
 - slice_by_status(model: ArchitectureModel, status: Status) -> ArchitectureModel
 - slice_for_artifact(model: ArchitectureModel, artifact_name: str) -> ArchitectureModel
- **F2.15** Typed dataclasses for all Architecture Model entity types and relationships [active] (src/architecture_model/core/types.py, 665 LOC)
- **F2.16** Validator: Check architecture model invariants [active] (src/architecture_model/core/validator.py, 518 LOC)
 - validate_model(model: ArchitectureModel, strict: bool) -> ValidationResult
- **F2.17** Extract an ArchitectureModel directly from source code analysis [active] (src/architecture_model/extract/from_code.py, 646 LOC)
 - extract_from_code(project_root: str | Path, config: ProjectConfig | None, manifest: dict | None) -> ArchitectureModel
- **F2.18** Reality Manifest Generator — modular package [active] (src/architecture_model/manifest/init.py, 58 LOC)
- **F2.19** Functional block definitions and processing [active] (src/architecture_model/manifest/blocks.py, 215 LOC)
 - process_block(root: Path, block_id: str, block_def: dict, sub_block_configs: list | None) -> BlockManifest
- **F2.20** Body Hints [active] (src/architecture_model/manifest/body_hints.py, 270 LOC)
 - classify_function(source: str, func_name: str) -> BodyComplexity
 - extract_body_hint(source: str, func_name: str, class_name: str | None) -> str
 - extract_file_hints(filepath: Path, include_private: bool) -> list[FunctionSignature]
- **F2.21** Terminal summary display for the reality manifest [active] (src/architecture_model/manifest/display.py, 51 LOC)
 - print_summary(manifest: dict[str, Any]) -> None

- **F2.22** Main manifest generation and caching logic [active] (src/architecture_model/manifest/generator.py, 168 LOC)
 - generate_manifest(project_root: Path, config: Optional[Any]) -> Manifest
 - load_or_generate_manifest(project_root: Path, output_dir: Path | None) -> dict[str, Any]
- **F2.23** Interface derivation from import analysis between scanned modules [active] (src/architecture_model/manifest/interfaces.py, 89 LOC)
 - derive_interfaces(modules: list[ModuleInfo], root: Path) -> list[InterfaceEdge]
- **F2.24** Project metrics computation from filesystem scanning [active] (src/architecture_model/manifest/metrics.py, 85 LOC)
 - compute_metrics(root: Path, config: Optional[Any]) -> MetricsResult
- **F2.25** AST helpers and file scanning for the reality manifest [active] (src/architecture_model/manifest/scanner.py, 496 LOC)
 - scan_file(root: Path, filepath: Path) -> ModuleInfo
- **F2.26** Manifest slicing for artifact context injection [active] (src/architecture_model/manifest/slicers.py, 579 LOC)
 - get_manifest_slice(manifest: Manifest | dict[str, Any], artifact_name: str) -> str
- **F2.27** Test Contract Analyzer — extracts behavioral specifications from test files [active] (src/architecture_model/manifest/test_analyzer.py, 501 LOC)
 - analyze_test_file(test_file: Path) -> TestAnalysisResult
 - extract_constants_from_contracts(contracts: list[TestContract]) -> list[Constant]
- **F2.28** Typed dataclasses for manifest generation outputs [active] (src/architecture_model/manifest/types.py, 219 LOC)
- **F2.29** Domain profiles for cross [dormant] (src/architecture_model/profiles/init.py, 1 LOC)
- **F2.30** Init [dormant] (src/architecture_model/profiles/builtins/init.py, 0 LOC)
- **F2.31** Domain profile schema and loading logic [active] (src/architecture_model/profiles/schema.py, 85 LOC)
 - load_profile(name_or_path: str) -> DomainProfile
- **F2.32** Init [dormant] (src/architecture_model/spec/init.py, 0 LOC)
- **F2.33** Shared utilities for architecture [dormant] (src/architecture_model/utils/init.py, 1 LOC)
- **F2.34** Shared file discovery and exclusion logic [active] (src/architecture_model/utils/discovery.py, 72 LOC)
 - is_excluded_dir(path: Path) -> bool
 - collect_py_files(directory: Path, recursive: bool, exclude_init: bool) -> list[Path]
 - discover_source_files(project_root: Path) -> list[Path]
 - discover_test_files(project_root: Path) -> list[Path]

F3: Core [ACTIVE]

Source: auto-discovered from src/architecture_model/core/

- **F3.1** Architecture Model Standard — universal machine [dormant] (src/architecture_model/init.py, 31 LOC)
- **F3.2** Package entry point: python [dormant] (src/architecture_model/main.py, 6 LOC)
- **F3.3** Init [dormant] (src/architecture_model/cli/init.py, 0 LOC)
- **F3.4** CLI for the Architecture Model Standard [active] (src/architecture_model/cli/main.py, 385 LOC)
 - main(argv: list[str] | None) -> int
- **F3.5** Configuration subpackage for the Architecture Model Standard [dormant] (src/architecture_model/config/init.py, 25 LOC)
- **F3.6** Configuration loader for the Architecture Model Standard [active] (src/architecture_model/config/loader.py, 558 LOC)

- load_config(root: Path) -> ProjectConfig
- discover_config(root: Path) -> tuple[ProjectConfig, DiscoveryReport]
- get_config(root: Path) -> ProjectConfig
- write_config(config: ProjectConfig, root: Path | None) -> Path
- **F3.7** Project configuration schema [active] (src/architecture_model/config/schema.py, 325 LOC)
- **F3.8** Init [dormant] (src/architecture_model/core/init.py, 0 LOC)
- **F3.9** Coverage analysis: compare architecture model against code reality (manifest) [active] (src/architecture_model/core/coverage.py, 268 LOC)
 - coverage_report(model: 'ArchitectureModel', manifest: dict) -> CoverageResult
- **F3.10** Complexity scoring and system identification for architecture decomposition [active] (src/architecture_model/core/decomposer.py, 659 LOC)
 - compute_complexity(comp: Component, model: ArchitectureModel) -> float
 - identify_systems(model: ArchitectureModel, manifest: dict) -> list[SystemCandidate]
 - auto_assign_f_blocks(model: ArchitectureModel, max_cluster_size: int) -> ArchitectureModel
 - decompose_model(model: ArchitectureModel, manifest: dict, output_dir: str) -> DecompositionResult
 - test_affinity_decompose(repo_path: Path) -> list[Subsystem]
- **F3.11** Differ: Compare two architecture model versions and produce a structured diff [active] (src/architecture_model/core/differ.py, 294 LOC)
 - diff_models(old_model: ArchitectureModel, new_model: ArchitectureModel) -> ModelDiff
- **F3.12** Merger: Merge manifest facts into an existing architecture model [active] (src/architecture_model/core/merger.py, 778 LOC)
 - merge_manifest(model: ArchitectureModel, manifest_path: str | Path, project_root: str | Path | None) -> ArchitectureModel
 - enrich_from_manifest(model: ArchitectureModel, manifest: dict) -> EnrichmentResult
 - compact_for_generation(model: ArchitectureModel) -> ArchitectureModel
 - compose_enriched_model(project_root: Path) -> ArchitectureModel
- **F3.13** Parser [active] (src/architecture_model/core/parser.py, 592 LOC)
 - load_model(path: str | Path) -> ArchitectureModel
 - validate_model_data(data: dict[str, Any]) -> list[str]
 - dump_model(model: ArchitectureModel) -> dict[str, Any]
 - save_model(model: ArchitectureModel, path: str | Path) -> None
- **F3.14** Slicer: Extract subsets of the architecture model [active] (src/architecture_model/core/slicer.py, 496 LOC)
 - slice_by_fblock(model: ArchitectureModel, f_block: str, include_relationships: bool) -> ArchitectureModel
 - slice_by_layer(model: ArchitectureModel, layer_id: str) -> ArchitectureModel
 - slice_by_status(model: ArchitectureModel, status: Status) -> ArchitectureModel
 - slice_for_artifact(model: ArchitectureModel, artifact_name: str) -> ArchitectureModel
- **F3.15** Typed dataclasses for all Architecture Model entity types and relationships [active] (src/architecture_model/core/types.py, 665 LOC)
- **F3.16** Validator: Check architecture model invariants [active] (src/architecture_model/core/validator.py, 518 LOC)
 - validate_model(model: ArchitectureModel, strict: bool) -> ValidationResult
- **F3.17** Extract an ArchitectureModel directly from source code analysis [active] (src/architecture_model/extract/from_code.py, 646 LOC)
 - extract_from_code(project_root: str | Path, config: ProjectConfig | None,

- manifest: dict | None) -> ArchitectureModel
- **F3.18** Reality Manifest Generator — modular package [active] (src/architecture_model/manifest/init.py, 58 LOC)
- **F3.19** Functional block definitions and processing [active] (src/architecture_model/manifest/blocks.py, 215 LOC)
 - process_block(root: Path, block_id: str, block_def: dict, sub_block_configs: list | None) -> BlockManifest
- **F3.20** Body Hints [active] (src/architecture_model/manifest/body_hints.py, 270 LOC)
 - classify_function(source: str, func_name: str) -> BodyComplexity
 - extract_body_hint(source: str, func_name: str, class_name: str | None) -> str
 - extract_file_hints(filepath: Path, include_private: bool) -> list[FunctionSignature]
- **F3.21** Terminal summary display for the reality manifest [active] (src/architecture_model/manifest/display.py, 51 LOC)
 - print_summary(manifest: dict[str, Any]) -> None
- **F3.22** Main manifest generation and caching logic [active] (src/architecture_model/manifest/generator.py, 168 LOC)
 - generate_manifest(project_root: Path, config: Optional[Any]) -> Manifest
 - load_or_generate_manifest(project_root: Path, output_dir: Path | None) -> dict[str, Any]
- **F3.23** Interface derivation from import analysis between scanned modules [active] (src/architecture_model/manifest/interfaces.py, 89 LOC)
 - derive_interfaces(modules: list[ModuleInfo], root: Path) -> list[InterfaceEdge]
- **F3.24** Project metrics computation from filesystem scanning [active] (src/architecture_model/manifest/metrics.py, 85 LOC)
 - compute_metrics(root: Path, config: Optional[Any]) -> MetricsResult
- **F3.25** AST helpers and file scanning for the reality manifest [active] (src/architecture_model/manifest/scanner.py, 496 LOC)
 - scan_file(root: Path, filepath: Path) -> ModuleInfo
- **F3.26** Manifest slicing for artifact context injection [active] (src/architecture_model/manifest/slicers.py, 579 LOC)
 - get_manifest_slice(manifest: Manifest | dict[str, Any], artifact_name: str) -> str
- **F3.27** Test Contract Analyzer — extracts behavioral specifications from test files [active] (src/architecture_model/manifest/test_analyzer.py, 501 LOC)
 - analyze_test_file(test_file: Path) -> TestAnalysisResult
 - extract_constants_from_contracts(contracts: list[TestContract]) -> list[Constant]
- **F3.28** Typed dataclasses for manifest generation outputs [active] (src/architecture_model/manifest/types.py, 219 LOC)
- **F3.29** Domain profiles for cross [dormant] (src/architecture_model/profiles/init.py, 1 LOC)
- **F3.30** Init [dormant] (src/architecture_model/profiles/builtins/init.py, 0 LOC)
- **F3.31** Domain profile schema and loading logic [active] (src/architecture_model/profiles/schema.py, 85 LOC)
 - load_profile(name_or_path: str) -> DomainProfile
- **F3.32** Init [dormant] (src/architecture_model/spec/init.py, 0 LOC)
- **F3.33** Shared utilities for architecture [dormant] (src/architecture_model/utils/init.py, 1 LOC)
- **F3.34** Shared file discovery and exclusion logic [active] (src/architecture_model/utils/discovery.py, 72 LOC)
 - is_excluded_dir(path: Path) -> bool
 - collect_py_files(directory: Path, recursive: bool, exclude_init: bool) -> list[Path]
 - discover_source_files(project_root: Path) -> list[Path]
 - discover_test_files(project_root: Path) -> list[Path]

F4: Extract [ACTIVE]

Source: auto-discovered from `src/architecture_model/extract/`

- **F4.1** Architecture Model Standard — universal machine [dormant] (`src/architecture_model/init.py`, 31 LOC)
- **F4.2** Package entry point: python [dormant] (`src/architecture_model/main.py`, 6 LOC)
- **F4.3** Init [dormant] (`src/architecture_model/cli/init.py`, 0 LOC)
- **F4.4** CLI for the Architecture Model Standard [active] (`src/architecture_model/cli/main.py`, 385 LOC)
 - `main(argv: list[str] | None) -> int`
- **F4.5** Configuration subpackage for the Architecture Model Standard [dormant] (`src/architecture_model/config/init.py`, 25 LOC)
- **F4.6** Configuration loader for the Architecture Model Standard [active] (`src/architecture_model/config/loader.py`, 558 LOC)
 - `load_config(root: Path) -> ProjectConfig`
 - `discover_config(root: Path) -> tuple[ProjectConfig, DiscoveryReport]`
 - `get_config(root: Path) -> ProjectConfig`
 - `write_config(config: ProjectConfig, root: Path | None) -> Path`
- **F4.7** Project configuration schema [active] (`src/architecture_model/config/schema.py`, 325 LOC)
- **F4.8** Init [dormant] (`src/architecture_model/core/init.py`, 0 LOC)
- **F4.9** Coverage analysis: compare architecture model against code reality (manifest) [active] (`src/architecture_model/core/coverage.py`, 268 LOC)
 - `coverage_report(model: 'ArchitectureModel', manifest: dict) -> CoverageResult`
- **F4.10** Complexity scoring and system identification for architecture decomposition [active] (`src/architecture_model/core/decomposer.py`, 659 LOC)
 - `compute_complexity(comp: Component, model: ArchitectureModel) -> float`
 - `identify_systems(model: ArchitectureModel, manifest: dict) -> list[SystemCandidate]`
 - `auto_assign_f_blocks(model: ArchitectureModel, max_cluster_size: int) -> ArchitectureModel`
 - `decompose_model(model: ArchitectureModel, manifest: dict, output_dir: str) -> DecompositionResult`
 - `test_affinity_decompose(repo_path: Path) -> list[Subsystem]`
- **F4.11** Differ: Compare two architecture model versions and produce a structured diff [active] (`src/architecture_model/core/differ.py`, 294 LOC)
 - `diff_models(old_model: ArchitectureModel, new_model: ArchitectureModel) -> ModelDiff`
- **F4.12** Merger: Merge manifest facts into an existing architecture model [active] (`src/architecture_model/core/merger.py`, 778 LOC)
 - `merge_manifest(model: ArchitectureModel, manifest_path: str | Path, project_root: str | Path | None) -> ArchitectureModel`
 - `enrich_from_manifest(model: ArchitectureModel, manifest: dict) -> EnrichmentResult`
 - `compact_for_generation(model: ArchitectureModel) -> ArchitectureModel`
 - `compose_enriched_model(project_root: Path) -> ArchitectureModel`
- **F4.13** Parser [active] (`src/architecture_model/core/parser.py`, 592 LOC)
 - `load_model(path: str | Path) -> ArchitectureModel`
 - `validate_model_data(data: dict[str, Any]) -> list[str]`
 - `dump_model(model: ArchitectureModel) -> dict[str, Any]`
 - `save_model(model: ArchitectureModel, path: str | Path) -> None`
- **F4.14** Slicer: Extract subsets of the architecture model [active] (`src/architecture_model/core/slicer.py`, 496 LOC)
 - `slice_by_fblock(model: ArchitectureModel, f_block: str, include_relationships: bool) -> ArchitectureModel`
 - `slice_by_layer(model: ArchitectureModel, layer_id: str) -> Architecture-`

- Model
 - slice_by_status(model: ArchitectureModel, status: Status) -> ArchitectureModel
 - slice_for_artifact(model: ArchitectureModel, artifact_name: str) -> ArchitectureModel
- **F4.15** Typed dataclasses for all Architecture Model entity types and relationships [active] (src/architecture_model/core/types.py, 665 LOC)
- **F4.16** Validator: Check architecture model invariants [active] (src/architecture_model/core/validator.py, 518 LOC)
 - validate_model(model: ArchitectureModel, strict: bool) -> ValidationResult
- **F4.17** Extract an ArchitectureModel directly from source code analysis [active] (src/architecture_model/extract/from_code.py, 646 LOC)
 - extract_from_code(project_root: str | Path, config: ProjectConfig | None, manifest: dict | None) -> ArchitectureModel
- **F4.18** Reality Manifest Generator — modular package [active] (src/architecture_model/manifest/init.py, 58 LOC)
- **F4.19** Functional block definitions and processing [active] (src/architecture_model/manifest/blocks.py, 215 LOC)
 - process_block(root: Path, block_id: str, block_def: dict, sub_block_configs: list | None) -> BlockManifest
- **F4.20** Body Hints [active] (src/architecture_model/manifest/body_hints.py, 270 LOC)
 - classify_function(source: str, func_name: str) -> BodyComplexity
 - extract_body_hint(source: str, func_name: str, class_name: str | None) -> str
 - extract_file_hints(filepath: Path, include_private: bool) -> list[FunctionSignature]
- **F4.21** Terminal summary display for the reality manifest [active] (src/architecture_model/manifest/display.py, 51 LOC)
 - print_summary(manifest: dict[str, Any]) -> None
- **F4.22** Main manifest generation and caching logic [active] (src/architecture_model/manifest/generator.py, 168 LOC)
 - generate_manifest(project_root: Path, config: Optional[Any]) -> Manifest
 - load_or_generate_manifest(project_root: Path, output_dir: Path | None) -> dict[str, Any]
- **F4.23** Interface derivation from import analysis between scanned modules [active] (src/architecture_model/manifest/interfaces.py, 89 LOC)
 - derive_interfaces(modules: list[ModuleInfo], root: Path) -> list[InterfaceEdge]
- **F4.24** Project metrics computation from filesystem scanning [active] (src/architecture_model/manifest/metrics.py, 85 LOC)
 - compute_metrics(root: Path, config: Optional[Any]) -> MetricsResult
- **F4.25** AST helpers and file scanning for the reality manifest [active] (src/architecture_model/manifest/scanner.py, 496 LOC)
 - scan_file(root: Path, filepath: Path) -> ModuleInfo
- **F4.26** Manifest slicing for artifact context injection [active] (src/architecture_model/manifest/slicers.py, 579 LOC)
 - get_manifest_slice(manifest: Manifest | dict[str, Any], artifact_name: str) -> str
- **F4.27** Test Contract Analyzer — extracts behavioral specifications from test files [active] (src/architecture_model/manifest/test_analyzer.py, 501 LOC)
 - analyze_test_file(test_file: Path) -> TestAnalysisResult
 - extract_constants_from_contracts(contracts: list[TestContract]) -> list[Constant]
- **F4.28** Typed dataclasses for manifest generation outputs [active] (src/architecture_model/manifest/types.py, 219 LOC)
- **F4.29** Domain profiles for cross [dormant] (src/architecture_model/profiles/init.py, 1 LOC)

- **F4.30** Init [dormant] (src/architecture_model/profiles/builtins/init.py, 0 LOC)
- **F4.31** Domain profile schema and loading logic [active] (src/architecture_model/profiles/schema.py, 85 LOC)
 - load_profile(name_or_path: str) -> DomainProfile
- **F4.32** Init [dormant] (src/architecture_model/spec/init.py, 0 LOC)
- **F4.33** Shared utilities for architecture [dormant] (src/architecture_model/utills/init.py, 1 LOC)
- **F4.34** Shared file discovery and exclusion logic [active] (src/architecture_model/utills/discovery.py, 72 LOC)
 - is_excluded_dir(path: Path) -> bool
 - collect_py_files(directory: Path, recursive: bool, exclude_init: bool) -> list[Path]
 - discover_source_files(project_root: Path) -> list[Path]
 - discover_test_files(project_root: Path) -> list[Path]

F5: Manifest [ACTIVE]

Source: Reality Manifest Generator — modular package.

- **F5.1** Architecture Model Standard — universal machine [dormant] (src/architecture_model/init.py, 31 LOC)
- **F5.2** Package entry point: python [dormant] (src/architecture_model/main.py, 6 LOC)
- **F5.3** Init [dormant] (src/architecture_model/cli/init.py, 0 LOC)
- **F5.4** CLI for the Architecture Model Standard [active] (src/architecture_model/cli/main.py, 385 LOC)
 - main(argv: list[str] | None) -> int
- **F5.5** Configuration subpackage for the Architecture Model Standard [dormant] (src/architecture_model/config/init.py, 25 LOC)
- **F5.6** Configuration loader for the Architecture Model Standard [active] (src/architecture_model/config/loader.py, 558 LOC)
 - load_config(root: Path) -> ProjectConfig
 - discover_config(root: Path) -> tuple[ProjectConfig, DiscoveryReport]
 - get_config(root: Path) -> ProjectConfig
 - write_config(config: ProjectConfig, root: Path | None) -> Path
- **F5.7** Project configuration schema [active] (src/architecture_model/config/schema.py, 325 LOC)
- **F5.8** Init [dormant] (src/architecture_model/core/init.py, 0 LOC)
- **F5.9** Coverage analysis: compare architecture model against code reality (manifest) [active] (src/architecture_model/core/coverage.py, 268 LOC)
 - coverage_report(model: 'ArchitectureModel', manifest: dict) -> CoverageResult
- **F5.10** Complexity scoring and system identification for architecture decomposition [active] (src/architecture_model/core/decomposer.py, 659 LOC)
 - compute_complexity(comp: Component, model: ArchitectureModel) -> float
 - identify_systems(model: ArchitectureModel, manifest: dict) -> list[SystemCandidate]
 - auto_assign_f_blocks(model: ArchitectureModel, max_cluster_size: int) -> ArchitectureModel
 - decompose_model(model: ArchitectureModel, manifest: dict, output_dir: str) -> DecompositionResult
 - test_affinity_decompose(repo_path: Path) -> list[Subsystem]
- **F5.11** Differ: Compare two architecture model versions and produce a structured diff [active] (src/architecture_model/core/differ.py, 294 LOC)
 - diff_models(old_model: ArchitectureModel, new_model: ArchitectureModel) -> ModelDiff
- **F5.12** Merger: Merge manifest facts into an existing architecture model [active] (src/architecture_model/core/merger.py, 778 LOC)
 - merge_manifest(model: ArchitectureModel, manifest_path: str | Path,

- project_root: str | Path | None) -> ArchitectureModel
 - enrich_from_manifest(model: ArchitectureModel, manifest: dict) -> EnrichmentResult
 - compact_for_generation(model: ArchitectureModel) -> ArchitectureModel
 - compose_enriched_model(project_root: Path) -> ArchitectureModel
- **F5.13** Parser [active] (src/architecture_model/core/parser.py, 592 LOC)
 - load_model(path: str | Path) -> ArchitectureModel
 - validate_model_data(data: dict[str, Any]) -> list[str]
 - dump_model(model: ArchitectureModel) -> dict[str, Any]
 - save_model(model: ArchitectureModel, path: str | Path) -> None
- **F5.14** Slicer: Extract subsets of the architecture model [active] (src/architecture_model/core/slicer.py, 496 LOC)
 - slice_by_fblock(model: ArchitectureModel, f_block: str, include_relationships: bool) -> ArchitectureModel
 - slice_by_layer(model: ArchitectureModel, layer_id: str) -> ArchitectureModel
 - slice_by_status(model: ArchitectureModel, status: Status) -> ArchitectureModel
 - slice_for_artifact(model: ArchitectureModel, artifact_name: str) -> ArchitectureModel
- **F5.15** Typed dataclasses for all Architecture Model entity types and relationships [active] (src/architecture_model/core/types.py, 665 LOC)
- **F5.16** Validator: Check architecture model invariants [active] (src/architecture_model/core/validator.py, 518 LOC)
 - validate_model(model: ArchitectureModel, strict: bool) -> ValidationResult
- **F5.17** Extract an ArchitectureModel directly from source code analysis [active] (src/architecture_model/extract/from_code.py, 646 LOC)
 - extract_from_code(project_root: str | Path, config: ProjectConfig | None, manifest: dict | None) -> ArchitectureModel
- **F5.18** Reality Manifest Generator — modular package [active] (src/architecture_model/manifest/init.py, 58 LOC)
- **F5.19** Functional block definitions and processing [active] (src/architecture_model/manifest/blocks.py, 215 LOC)
 - process_block(root: Path, block_id: str, block_def: dict, sub_block_configs: list | None) -> BlockManifest
- **F5.20** Body Hints [active] (src/architecture_model/manifest/body_hints.py, 270 LOC)
 - classify_function(source: str, func_name: str) -> BodyComplexity
 - extract_body_hint(source: str, func_name: str, class_name: str | None) -> str
 - extract_file_hints(filepath: Path, include_private: bool) -> list[FunctionSignature]
- **F5.21** Terminal summary display for the reality manifest [active] (src/architecture_model/manifest/display.py, 51 LOC)
 - print_summary(manifest: dict[str, Any]) -> None
- **F5.22** Main manifest generation and caching logic [active] (src/architecture_model/manifest/generator.py, 168 LOC)
 - generate_manifest(project_root: Path, config: Optional[Any]) -> Manifest
 - load_or_generate_manifest(project_root: Path, output_dir: Path | None) -> dict[str, Any]
- **F5.23** Interface derivation from import analysis between scanned modules [active] (src/architecture_model/manifest/interfaces.py, 89 LOC)
 - derive_interfaces(modules: list[ModuleInfo], root: Path) -> list[InterfaceEdge]
- **F5.24** Project metrics computation from filesystem scanning [active] (src/architecture_model/manifest/metrics.py, 85 LOC)
 - compute_metrics(root: Path, config: Optional[Any]) -> MetricsResult

- **F5.25** AST helpers and file scanning for the reality manifest [active] (src/architecture_model/manifest/scanner.py, 496 LOC)
 - scan_file(root: Path, filepath: Path) -> ModuleInfo
- **F5.26** Manifest slicing for artifact context injection [active] (src/architecture_model/manifest/slicers.py, 579 LOC)
 - get_manifest_slice(manifest: Manifest | dict[str, Any], artifact_name: str) -> str
- **F5.27** Test Contract Analyzer — extracts behavioral specifications from test files [active] (src/architecture_model/manifest/test_analyzer.py, 501 LOC)
 - analyze_test_file(test_file: Path) -> TestAnalysisResult
 - extract_constants_from_contracts(contracts: list[TestContract]) -> list[Constant]
- **F5.28** Typed dataclasses for manifest generation outputs [active] (src/architecture_model/manifest/types.py, 219 LOC)
- **F5.29** Domain profiles for cross [dormant] (src/architecture_model/profiles/init.py, 1 LOC)
- **F5.30** Init [dormant] (src/architecture_model/profiles/builtins/init.py, 0 LOC)
- **F5.31** Domain profile schema and loading logic [active] (src/architecture_model/profiles/schema.py, 85 LOC)
 - load_profile(name_or_path: str) -> DomainProfile
- **F5.32** Init [dormant] (src/architecture_model/spec/init.py, 0 LOC)
- **F5.33** Shared utilities for architecture [dormant] (src/architecture_model/utils/init.py, 1 LOC)
- **F5.34** Shared file discovery and exclusion logic [active] (src/architecture_model/utils/discovery.py, 72 LOC)
 - is_excluded_dir(path: Path) -> bool
 - collect_py_files(directory: Path, recursive: bool, exclude_init: bool) -> list[Path]
 - discover_source_files(project_root: Path) -> list[Path]
 - discover_test_files(project_root: Path) -> list[Path]

F6: Profiles [ACTIVE]

Source: Domain profiles for cross-domain architecture modeling.

- **F6.1** Architecture Model Standard — universal machine [dormant] (src/architecture_model/init.py, 31 LOC)
- **F6.2** Package entry point: python [dormant] (src/architecture_model/main.py, 6 LOC)
- **F6.3** Init [dormant] (src/architecture_model/cli/init.py, 0 LOC)
- **F6.4** CLI for the Architecture Model Standard [active] (src/architecture_model/cli/main.py, 385 LOC)
 - main(argv: list[str] | None) -> int
- **F6.5** Configuration subpackage for the Architecture Model Standard [dormant] (src/architecture_model/config/init.py, 25 LOC)
- **F6.6** Configuration loader for the Architecture Model Standard [active] (src/architecture_model/config/loader.py, 558 LOC)
 - load_config(root: Path) -> ProjectConfig
 - discover_config(root: Path) -> tuple[ProjectConfig, DiscoveryReport]
 - get_config(root: Path) -> ProjectConfig
 - write_config(config: ProjectConfig, root: Path | None) -> Path
- **F6.7** Project configuration schema [active] (src/architecture_model/config/schema.py, 325 LOC)
- **F6.8** Init [dormant] (src/architecture_model/core/init.py, 0 LOC)
- **F6.9** Coverage analysis: compare architecture model against code reality (manifest) [active] (src/architecture_model/core/coverage.py, 268 LOC)
 - coverage_report(model: 'ArchitectureModel', manifest: dict) -> CoverageResult
- **F6.10** Complexity scoring and system identification for architecture decomposition [active]

- (src/architecture_model/core/decomposer.py, 659 LOC)
 - compute_complexity(comp: Component, model: ArchitectureModel) -> float
 - identify_systems(model: ArchitectureModel, manifest: dict) -> list[SystemCandidate]
 - auto_assign_f_blocks(model: ArchitectureModel, max_cluster_size: int) -> ArchitectureModel
 - decompose_model(model: ArchitectureModel, manifest: dict, output_dir: str) -> DecompositionResult
 - test_affinity_decompose(repo_path: Path) -> list[Subsystem]
- **F6.11** Differ: Compare two architecture model versions and produce a structured diff [active] (src/architecture_model/core/differ.py, 294 LOC)
 - diff_models(old_model: ArchitectureModel, new_model: ArchitectureModel) -> ModelDiff
- **F6.12** Merger: Merge manifest facts into an existing architecture model [active] (src/architecture_model/core/merger.py, 778 LOC)
 - merge_manifest(model: ArchitectureModel, manifest_path: str | Path, project_root: str | Path | None) -> ArchitectureModel
 - enrich_from_manifest(model: ArchitectureModel, manifest: dict) -> EnrichmentResult
 - compact_for_generation(model: ArchitectureModel) -> ArchitectureModel
 - compose_enriched_model(project_root: Path) -> ArchitectureModel
- **F6.13** Parser [active] (src/architecture_model/core/parser.py, 592 LOC)
 - load_model(path: str | Path) -> ArchitectureModel
 - validate_model_data(data: dict[str, Any]) -> list[str]
 - dump_model(model: ArchitectureModel) -> dict[str, Any]
 - save_model(model: ArchitectureModel, path: str | Path) -> None
- **F6.14** Slicer: Extract subsets of the architecture model [active] (src/architecture_model/core/slicer.py, 496 LOC)
 - slice_by_fblock(model: ArchitectureModel, f_block: str, include_relationships: bool) -> ArchitectureModel
 - slice_by_layer(model: ArchitectureModel, layer_id: str) -> ArchitectureModel
 - slice_by_status(model: ArchitectureModel, status: Status) -> ArchitectureModel
 - slice_for_artifact(model: ArchitectureModel, artifact_name: str) -> ArchitectureModel
- **F6.15** Typed dataclasses for all Architecture Model entity types and relationships [active] (src/architecture_model/core/types.py, 665 LOC)
- **F6.16** Validator: Check architecture model invariants [active] (src/architecture_model/core/validator.py, 518 LOC)
 - validate_model(model: ArchitectureModel, strict: bool) -> ValidationResult
- **F6.17** Extract an ArchitectureModel directly from source code analysis [active] (src/architecture_model/extract/from_code.py, 646 LOC)
 - extract_from_code(project_root: str | Path, config: ProjectConfig | None, manifest: dict | None) -> ArchitectureModel
- **F6.18** Reality Manifest Generator — modular package [active] (src/architecture_model/manifest/init.py, 58 LOC)
- **F6.19** Functional block definitions and processing [active] (src/architecture_model/manifest/blocks.py, 215 LOC)
 - process_block(root: Path, block_id: str, block_def: dict, sub_block_configs: list | None) -> BlockManifest
- **F6.20** Body Hints [active] (src/architecture_model/manifest/body_hints.py, 270 LOC)
 - classify_function(source: str, func_name: str) -> BodyComplexity
 - extract_body_hint(source: str, func_name: str, class_name: str | None) -> str

- extract_file_hints(filepath: Path, include_private: bool) -> list[FunctionSignature]
- **F6.21** Terminal summary display for the reality manifest [active] (src/architecture_model/manifest/display.py, 51 LOC)
 - print_summary(manifest: dict[str, Any]) -> None
- **F6.22** Main manifest generation and caching logic [active] (src/architecture_model/manifest/generator.py, 168 LOC)
 - generate_manifest(project_root: Path, config: Optional[Any]) -> Manifest
 - load_or_generate_manifest(project_root: Path, output_dir: Path | None) -> dict[str, Any]
- **F6.23** Interface derivation from import analysis between scanned modules [active] (src/architecture_model/manifest/interfaces.py, 89 LOC)
 - derive_interfaces(modules: list[ModuleInfo], root: Path) -> list[InterfaceEdge]
- **F6.24** Project metrics computation from filesystem scanning [active] (src/architecture_model/manifest/metrics.py, 85 LOC)
 - compute_metrics(root: Path, config: Optional[Any]) -> MetricsResult
- **F6.25** AST helpers and file scanning for the reality manifest [active] (src/architecture_model/manifest/scanner.py, 496 LOC)
 - scan_file(root: Path, filepath: Path) -> ModuleInfo
- **F6.26** Manifest slicing for artifact context injection [active] (src/architecture_model/manifest/slicers.py, 579 LOC)
 - get_manifest_slice(manifest: Manifest | dict[str, Any], artifact_name: str) -> str
- **F6.27** Test Contract Analyzer — extracts behavioral specifications from test files [active] (src/architecture_model/manifest/test_analyzer.py, 501 LOC)
 - analyze_test_file(test_file: Path) -> TestAnalysisResult
 - extract_constants_from_contracts(contracts: list[TestContract]) -> list[Constant]
- **F6.28** Typed dataclasses for manifest generation outputs [active] (src/architecture_model/manifest/types.py, 219 LOC)
- **F6.29** Domain profiles for cross [dormant] (src/architecture_model/profiles/init.py, 1 LOC)
- **F6.30** Init [dormant] (src/architecture_model/profiles/builtins/init.py, 0 LOC)
- **F6.31** Domain profile schema and loading logic [active] (src/architecture_model/profiles/schema.py, 85 LOC)
 - load_profile(name_or_path: str) -> DomainProfile
- **F6.32** Init [dormant] (src/architecture_model/spec/init.py, 0 LOC)
- **F6.33** Shared utilities for architecture [dormant] (src/architecture_model/utils/init.py, 1 LOC)
- **F6.34** Shared file discovery and exclusion logic [active] (src/architecture_model/utils/discovery.py, 72 LOC)
 - is_excluded_dir(path: Path) -> bool
 - collect_py_files(directory: Path, recursive: bool, exclude_init: bool) -> list[Path]
 - discover_source_files(project_root: Path) -> list[Path]
 - discover_test_files(project_root: Path) -> list[Path]

F6.A: Builtins [dormant]

- **F6.30** Init (src/architecture_model/profiles/builtins/init.py, 0 LOC)

F6.misc: Ungrouped [active]

- **F6.1** Architecture Model Standard — universal machine (src/architecture_model/init.py, 31 LOC)
- **F6.2** Package entry point: python (src/architecture_model/main.py, 6 LOC)
- **F6.3** Init (src/architecture_model/cli/init.py, 0 LOC)
- **F6.4** CLI for the Architecture Model Standard (src/architecture_model/cli/main.py, 385 LOC)

- main(argv: list[str] | None) -> int
- **F6.5** Configuration subpackage for the Architecture Model Standard (src/architecture_model/config/init.py, 25 LOC)
- **F6.6** Configuration loader for the Architecture Model Standard (src/architecture_model/config/loader.py, 558 LOC)
 - load_config(root: Path) -> ProjectConfig
 - discover_config(root: Path) -> tuple[ProjectConfig, DiscoveryReport]
 - get_config(root: Path) -> ProjectConfig
- **F6.7** Project configuration schema (src/architecture_model/config/schema.py, 325 LOC)
- **F6.8** Init (src/architecture_model/core/init.py, 0 LOC)
- **F6.9** Coverage analysis: compare architecture model against code reality (manifest) (src/architecture_model/core/coverage.py, 268 LOC)
 - coverage_report(model: 'ArchitectureModel', manifest: dict) -> CoverageResult
- **F6.10** Complexity scoring and system identification for architecture decomposition (src/architecture_model/core/decomposition.py, 659 LOC)
 - compute_complexity(comp: Component, model: ArchitectureModel) -> float
 - identify_systems(model: ArchitectureModel, manifest: dict) -> list[SystemCandidate]
 - auto_assign_f_blocks(model: ArchitectureModel, max_cluster_size: int) -> ArchitectureModel
- **F6.11** Differ: Compare two architecture model versions and produce a structured diff (src/architecture_model/core/differ.py, 294 LOC)
 - diff_models(old_model: ArchitectureModel, new_model: ArchitectureModel) -> ModelDiff
- **F6.12** Merger: Merge manifest facts into an existing architecture model (src/architecture_model/core/merger.py, 778 LOC)
 - merge_manifest(model: ArchitectureModel, manifest_path: str | Path, project_root: str | Path | None) -> ArchitectureModel
 - enrich_from_manifest(model: ArchitectureModel, manifest: dict) -> EnrichmentResult
 - compact_for_generation(model: ArchitectureModel) -> ArchitectureModel
- **F6.13** Parser (src/architecture_model/core/parser.py, 592 LOC)
 - load_model(path: str | Path) -> ArchitectureModel
 - validate_model_data(data: dict[str, Any]) -> list[str]
 - dump_model(model: ArchitectureModel) -> dict[str, Any]
- **F6.14** Slicer: Extract subsets of the architecture model (src/architecture_model/core/slicer.py, 496 LOC)
 - slice_by_fblock(model: ArchitectureModel, f_block: str, include_relationships: bool) -> ArchitectureModel
 - slice_by_layer(model: ArchitectureModel, layer_id: str) -> ArchitectureModel
 - slice_by_status(model: ArchitectureModel, status: Status) -> ArchitectureModel
- **F6.15** Typed dataclasses for all Architecture Model entity types and relationships (src/architecture_model/core/types.py, 665 LOC)
- **F6.16** Validator: Check architecture model invariants (src/architecture_model/core/validator.py, 518 LOC)
 - validate_model(model: ArchitectureModel, strict: bool) -> ValidationResult
- **F6.17** Extract an ArchitectureModel directly from source code analysis (src/architecture_model/extract/from_code.py, 646 LOC)
 - extract_from_code(project_root: str | Path, config: ProjectConfig | None, manifest: dict | None) -> ArchitectureModel
- **F6.18** Reality Manifest Generator — modular package (src/architecture_model/manifest/init.py, 58 LOC)

- **F6.19** Functional block definitions and processing (src/architecture_model/manifest/blocks.py, 215 LOC)
 - process_block(root: Path, block_id: str, block_def: dict, sub_block_configs: list | None) -> BlockManifest
- **F6.20** Body Hints (src/architecture_model/manifest/body_hints.py, 270 LOC)
 - classify_function(source: str, func_name: str) -> BodyComplexity
 - extract_body_hint(source: str, func_name: str, class_name: str | None) -> str
 - extract_file_hints(filepath: Path, include_private: bool) -> list[FunctionSignature]
- **F6.21** Terminal summary display for the reality manifest (src/architecture_model/manifest/display.py, 51 LOC)
 - print_summary(manifest: dict[str, Any]) -> None
- **F6.22** Main manifest generation and caching logic (src/architecture_model/manifest/generator.py, 168 LOC)
 - generate_manifest(project_root: Path, config: Optional[Any]) -> Manifest
 - load_or_generate_manifest(project_root: Path, output_dir: Path | None) -> dict[str, Any]
- **F6.23** Interface derivation from import analysis between scanned modules (src/architecture_model/manifest/interfaces.py, 89 LOC)
 - derive_interfaces(modules: list[ModuleInfo], root: Path) -> list[InterfaceEdge]
- **F6.24** Project metrics computation from filesystem scanning (src/architecture_model/manifest/metrics.py, 85 LOC)
 - compute_metrics(root: Path, config: Optional[Any]) -> MetricsResult
- **F6.25** AST helpers and file scanning for the reality manifest (src/architecture_model/manifest/scanner.py, 496 LOC)
 - scan_file(root: Path, filepath: Path) -> ModuleInfo
- **F6.26** Manifest slicing for artifact context injection (src/architecture_model/manifest/slicers.py, 579 LOC)
 - get_manifest_slice(manifest: Manifest | dict[str, Any], artifact_name: str) -> str
- **F6.27** Test Contract Analyzer — extracts behavioral specifications from test files (src/architecture_model/manifest/test_analyzer.py, 501 LOC)
 - analyze_test_file(test_file: Path) -> TestAnalysisResult
 - extract_constants_from_contracts(contracts: list[TestContract]) -> list[Constant]
- **F6.28** Typed dataclasses for manifest generation outputs (src/architecture_model/manifest/types.py, 219 LOC)
- **F6.29** Domain profiles for cross (src/architecture_model/profiles/init.py, 1 LOC)
- **F6.31** Domain profile schema and loading logic (src/architecture_model/profiles/schema.py, 85 LOC)
 - load_profile(name_or_path: str) -> DomainProfile
- **F6.32** Init (src/architecture_model/spec/init.py, 0 LOC)
- **F6.33** Shared utilities for architecture (src/architecture_model/utils/init.py, 1 LOC)
- **F6.34** Shared file discovery and exclusion logic (src/architecture_model/utils/discovery.py, 72 LOC)
 - is_excluded_dir(path: Path) -> bool
 - collect_py_files(directory: Path, recursive: bool, exclude_init: bool) -> list[Path]
 - discover_source_files(project_root: Path) -> list[Path]

F7: Spec [ACTIVE]

Source: auto-discovered from src/architecture_model/spec/

- **F7.1** Architecture Model Standard — universal machine [dormant] (src/architecture_model/init.py, 31 LOC)

- **F7.2** Package entry point: python [dormant] (src/architecture_model/**main.py**, 6 LOC)
- **F7.3** Init [dormant] (src/architecture_model/cli/**init.py**, 0 LOC)
- **F7.4** CLI for the Architecture Model Standard [active] (src/architecture_model/cli/main.py, 385 LOC)
 - main(argv: list[str] | None) -> int
- **F7.5** Configuration subpackage for the Architecture Model Standard [dormant] (src/architecture_model/config/**init.py**, 25 LOC)
- **F7.6** Configuration loader for the Architecture Model Standard [active] (src/architecture_model/config/loader.py, 558 LOC)
 - load_config(root: Path) -> ProjectConfig
 - discover_config(root: Path) -> tuple[ProjectConfig, DiscoveryReport]
 - get_config(root: Path) -> ProjectConfig
 - write_config(config: ProjectConfig, root: Path | None) -> Path
- **F7.7** Project configuration schema [active] (src/architecture_model/config/schema.py, 325 LOC)
- **F7.8** Init [dormant] (src/architecture_model/core/**init.py**, 0 LOC)
- **F7.9** Coverage analysis: compare architecture model against code reality (manifest) [active] (src/architecture_model/core/coverage.py, 268 LOC)
 - coverage_report(model: 'ArchitectureModel', manifest: dict) -> CoverageResult
- **F7.10** Complexity scoring and system identification for architecture decomposition [active] (src/architecture_model/core/decomposer.py, 659 LOC)
 - compute_complexity(comp: Component, model: ArchitectureModel) -> float
 - identify_systems(model: ArchitectureModel, manifest: dict) -> list[SystemCandidate]
 - auto_assign_f_blocks(model: ArchitectureModel, max_cluster_size: int) -> ArchitectureModel
 - decompose_model(model: ArchitectureModel, manifest: dict, output_dir: str) -> DecompositionResult
 - test_affinity_decompose(repo_path: Path) -> list[Subsystem]
- **F7.11** Differ: Compare two architecture model versions and produce a structured diff [active] (src/architecture_model/core/differ.py, 294 LOC)
 - diff_models(old_model: ArchitectureModel, new_model: ArchitectureModel) -> ModelDiff
- **F7.12** Merger: Merge manifest facts into an existing architecture model [active] (src/architecture_model/core/merger.py, 778 LOC)
 - merge_manifest(model: ArchitectureModel, manifest_path: str | Path, project_root: str | Path | None) -> ArchitectureModel
 - enrich_from_manifest(model: ArchitectureModel, manifest: dict) -> EnrichmentResult
 - compact_for_generation(model: ArchitectureModel) -> ArchitectureModel
 - compose_enriched_model(project_root: Path) -> ArchitectureModel
- **F7.13** Parser [active] (src/architecture_model/core/parser.py, 592 LOC)
 - load_model(path: str | Path) -> ArchitectureModel
 - validate_model_data(data: dict[str, Any]) -> list[str]
 - dump_model(model: ArchitectureModel) -> dict[str, Any]
 - save_model(model: ArchitectureModel, path: str | Path) -> None
- **F7.14** Slicer: Extract subsets of the architecture model [active] (src/architecture_model/core/slicer.py, 496 LOC)
 - slice_by_fblock(model: ArchitectureModel, f_block: str, include_relationships: bool) -> ArchitectureModel
 - slice_by_layer(model: ArchitectureModel, layer_id: str) -> ArchitectureModel
 - slice_by_status(model: ArchitectureModel, status: Status) -> ArchitectureModel
 - slice_for_artifact(model: ArchitectureModel, artifact_name: str) -> ArchitectureModel

- **F7.15** Typed dataclasses for all Architecture Model entity types and relationships [active] (src/architecture_model/core/types.py, 665 LOC)
- **F7.16** Validator: Check architecture model invariants [active] (src/architecture_model/core/validator.py, 518 LOC)
 - validate_model(model: ArchitectureModel, strict: bool) -> ValidationResult
- **F7.17** Extract an ArchitectureModel directly from source code analysis [active] (src/architecture_model/extract/from_code.py, 646 LOC)
 - extract_from_code(project_root: str | Path, config: ProjectConfig | None, manifest: dict | None) -> ArchitectureModel
- **F7.18** Reality Manifest Generator — modular package [active] (src/architecture_model/manifest/init.py, 58 LOC)
- **F7.19** Functional block definitions and processing [active] (src/architecture_model/manifest/blocks.py, 215 LOC)
 - process_block(root: Path, block_id: str, block_def: dict, sub_block_configs: list | None) -> BlockManifest
- **F7.20** Body Hints [active] (src/architecture_model/manifest/body_hints.py, 270 LOC)
 - classify_function(source: str, func_name: str) -> BodyComplexity
 - extract_body_hint(source: str, func_name: str, class_name: str | None) -> str
 - extract_file_hints(filepath: Path, include_private: bool) -> list[FunctionSignature]
- **F7.21** Terminal summary display for the reality manifest [active] (src/architecture_model/manifest/display.py, 51 LOC)
 - print_summary(manifest: dict[str, Any]) -> None
- **F7.22** Main manifest generation and caching logic [active] (src/architecture_model/manifest/generator.py, 168 LOC)
 - generate_manifest(project_root: Path, config: Optional[Any]) -> Manifest
 - load_or_generate_manifest(project_root: Path, output_dir: Path | None) -> dict[str, Any]
- **F7.23** Interface derivation from import analysis between scanned modules [active] (src/architecture_model/manifest/interfaces.py, 89 LOC)
 - derive_interfaces(modules: list[ModuleInfo], root: Path) -> list[InterfaceEdge]
- **F7.24** Project metrics computation from filesystem scanning [active] (src/architecture_model/manifest/metrics.py, 85 LOC)
 - compute_metrics(root: Path, config: Optional[Any]) -> MetricsResult
- **F7.25** AST helpers and file scanning for the reality manifest [active] (src/architecture_model/manifest/scanner.py, 496 LOC)
 - scan_file(root: Path, filepath: Path) -> ModuleInfo
- **F7.26** Manifest slicing for artifact context injection [active] (src/architecture_model/manifest/slicers.py, 579 LOC)
 - get_manifest_slice(manifest: Manifest | dict[str, Any], artifact_name: str) -> str
- **F7.27** Test Contract Analyzer — extracts behavioral specifications from test files [active] (src/architecture_model/manifest/test_analyzer.py, 501 LOC)
 - analyze_test_file(test_file: Path) -> TestAnalysisResult
 - extract_constants_from_contracts(contracts: list[TestContract]) -> list[Constant]
- **F7.28** Typed dataclasses for manifest generation outputs [active] (src/architecture_model/manifest/types.py, 219 LOC)
- **F7.29** Domain profiles for cross [dormant] (src/architecture_model/profiles/init.py, 1 LOC)
- **F7.30** Init [dormant] (src/architecture_model/profiles/builtins/init.py, 0 LOC)
- **F7.31** Domain profile schema and loading logic [active] (src/architecture_model/profiles/schema.py, 85 LOC)
 - load_profile(name_or_path: str) -> DomainProfile
- **F7.32** Init [dormant] (src/architecture_model/spec/init.py, 0 LOC)

- **F7.33** Shared utilities for architecture [dormant] (src/architecture_model/utis/**init**.py, 1 LOC)
- **F7.34** Shared file discovery and exclusion logic [active] (src/architecture_model/utis/discovery.py, 72 LOC)
 - is_excluded_dir(path: Path) -> bool
 - collect_py_files(directory: Path, recursive: bool, exclude_init: bool) -> list[Path]
 - discover_source_files(project_root: Path) -> list[Path]
 - discover_test_files(project_root: Path) -> list[Path]

F8: Utils [ACTIVE]

Source: Shared utilities for architecture-model-standard.

- **F8.1** Architecture Model Standard — universal machine [dormant] (src/architecture_model/**init**.py, 31 LOC)
- **F8.2** Package entry point: python [dormant] (src/architecture_model/**main**.py, 6 LOC)
- **F8.3** Init [dormant] (src/architecture_model/cli/**init**.py, 0 LOC)
- **F8.4** CLI for the Architecture Model Standard [active] (src/architecture_model/cli/main.py, 385 LOC)
 - main(argv: list[str] | None) -> int
- **F8.5** Configuration subpackage for the Architecture Model Standard [dormant] (src/architecture_model/config/**init**.py, 25 LOC)
- **F8.6** Configuration loader for the Architecture Model Standard [active] (src/architecture_model/config/loader.py, 558 LOC)
 - load_config(root: Path) -> ProjectConfig
 - discover_config(root: Path) -> tuple[ProjectConfig, DiscoveryReport]
 - get_config(root: Path) -> ProjectConfig
 - write_config(config: ProjectConfig, root: Path | None) -> Path
- **F8.7** Project configuration schema [active] (src/architecture_model/config/schema.py, 325 LOC)
- **F8.8** Init [dormant] (src/architecture_model/core/**init**.py, 0 LOC)
- **F8.9** Coverage analysis: compare architecture model against code reality (manifest) [active] (src/architecture_model/core/coverage.py, 268 LOC)
 - coverage_report(model: 'ArchitectureModel', manifest: dict) -> CoverageResult
- **F8.10** Complexity scoring and system identification for architecture decomposition [active] (src/architecture_model/core/decomposer.py, 659 LOC)
 - compute_complexity(comp: Component, model: ArchitectureModel) -> float
 - identify_systems(model: ArchitectureModel, manifest: dict) -> list[SystemCandidate]
 - auto_assign_f_blocks(model: ArchitectureModel, max_cluster_size: int) -> ArchitectureModel
 - decompose_model(model: ArchitectureModel, manifest: dict, output_dir: str) -> DecompositionResult
 - test_affinity_decompose(repo_path: Path) -> list[Subsystem]
- **F8.11** Differ: Compare two architecture model versions and produce a structured diff [active] (src/architecture_model/core/differ.py, 294 LOC)
 - diff_models(old_model: ArchitectureModel, new_model: ArchitectureModel) -> ModelDiff
- **F8.12** Merger: Merge manifest facts into an existing architecture model [active] (src/architecture_model/core/merger.py, 778 LOC)
 - merge_manifest(model: ArchitectureModel, manifest_path: str | Path, project_root: str | Path | None) -> ArchitectureModel
 - enrich_from_manifest(model: ArchitectureModel, manifest: dict) -> EnrichmentResult
 - compact_for_generation(model: ArchitectureModel) -> ArchitectureModel
 - compose_enriched_model(project_root: Path) -> ArchitectureModel

- **F8.13** Parser [active] (src/architecture_model/core/parser.py, 592 LOC)
 - load_model(path: str | Path) -> ArchitectureModel
 - validate_model_data(data: dict[str, Any]) -> list[str]
 - dump_model(model: ArchitectureModel) -> dict[str, Any]
 - save_model(model: ArchitectureModel, path: str | Path) -> None
- **F8.14** Slicer: Extract subsets of the architecture model [active] (src/architecture_model/core/slicer.py, 496 LOC)
 - slice_by_fblock(model: ArchitectureModel, f_block: str, include_relationships: bool) -> ArchitectureModel
 - slice_by_layer(model: ArchitectureModel, layer_id: str) -> ArchitectureModel
 - slice_by_status(model: ArchitectureModel, status: Status) -> ArchitectureModel
 - slice_for_artifact(model: ArchitectureModel, artifact_name: str) -> ArchitectureModel
- **F8.15** Typed dataclasses for all Architecture Model entity types and relationships [active] (src/architecture_model/core/types.py, 665 LOC)
- **F8.16** Validator: Check architecture model invariants [active] (src/architecture_model/core/validator.py, 518 LOC)
 - validate_model(model: ArchitectureModel, strict: bool) -> ValidationResult
- **F8.17** Extract an ArchitectureModel directly from source code analysis [active] (src/architecture_model/extract/from_code.py, 646 LOC)
 - extract_from_code(project_root: str | Path, config: ProjectConfig | None, manifest: dict | None) -> ArchitectureModel
- **F8.18** Reality Manifest Generator — modular package [active] (src/architecture_model/manifest/init.py, 58 LOC)
- **F8.19** Functional block definitions and processing [active] (src/architecture_model/manifest/blocks.py, 215 LOC)
 - process_block(root: Path, block_id: str, block_def: dict, sub_block_configs: list | None) -> BlockManifest
- **F8.20** Body Hints [active] (src/architecture_model/manifest/body_hints.py, 270 LOC)
 - classify_function(source: str, func_name: str) -> BodyComplexity
 - extract_body_hint(source: str, func_name: str, class_name: str | None) -> str
 - extract_file_hints(filepath: Path, include_private: bool) -> list[FunctionSignature]
- **F8.21** Terminal summary display for the reality manifest [active] (src/architecture_model/manifest/display.py, 51 LOC)
 - print_summary(manifest: dict[str, Any]) -> None
- **F8.22** Main manifest generation and caching logic [active] (src/architecture_model/manifest/generator.py, 168 LOC)
 - generate_manifest(project_root: Path, config: Optional[Any]) -> Manifest
 - load_or_generate_manifest(project_root: Path, output_dir: Path | None) -> dict[str, Any]
- **F8.23** Interface derivation from import analysis between scanned modules [active] (src/architecture_model/manifest/interfaces.py, 89 LOC)
 - derive_interfaces(modules: list[ModuleInfo], root: Path) -> list[InterfaceEdge]
- **F8.24** Project metrics computation from filesystem scanning [active] (src/architecture_model/manifest/metrics.py, 85 LOC)
 - compute_metrics(root: Path, config: Optional[Any]) -> MetricsResult
- **F8.25** AST helpers and file scanning for the reality manifest [active] (src/architecture_model/manifest/scanner.py, 496 LOC)
 - scan_file(root: Path, filepath: Path) -> ModuleInfo
- **F8.26** Manifest slicing for artifact context injection [active] (src/architecture_model/manifest/slicers.py, 579 LOC)

- get_manifest_slice(manifest: Manifest | dict[str, Any], artifact_name: str) → str
- **F8.27** Test Contract Analyzer — extracts behavioral specifications from test files [active] (src/architecture_model/manifest/test_analyzer.py, 501 LOC)
 - analyze_test_file(test_file: Path) → TestAnalysisResult
 - extract_constants_from_contracts(contracts: list[TestContract]) → list[Constant]
- **F8.28** Typed dataclasses for manifest generation outputs [active] (src/architecture_model/manifest/types.py, 219 LOC)
- **F8.29** Domain profiles for cross [dormant] (src/architecture_model/profiles/init.py, 1 LOC)
- **F8.30** Init [dormant] (src/architecture_model/profiles/builtins/init.py, 0 LOC)
- **F8.31** Domain profile schema and loading logic [active] (src/architecture_model/profiles/schema.py, 85 LOC)
 - load_profile(name_or_path: str) → DomainProfile
- **F8.32** Init [dormant] (src/architecture_model/spec/init.py, 0 LOC)
- **F8.33** Shared utilities for architecture [dormant] (src/architecture_model/utils/init.py, 1 LOC)
- **F8.34** Shared file discovery and exclusion logic [active] (src/architecture_model/utils/discovery.py, 72 LOC)
 - is_excluded_dir(path: Path) → bool
 - collect_py_files(directory: Path, recursive: bool, exclude_init: bool) → list[Path]
 - discover_source_files(project_root: Path) → list[Path]
 - discover_test_files(project_root: Path) → list[Path]

Modules

src/architecture_model/init.py [DORMANT] (31 LOC)

Architecture Model Standard — universal machine-readable architectural spine.

src/architecture_model/main.py [DORMANT] (6 LOC)

Package entry point: python -m architecture_model

src/architecture_model/cli/init.py [DORMANT] (0 LOC)

src/architecture_model/cli/main.py [ACTIVE] (385 LOC)

CLI for the Architecture Model Standard.

- main(argv: list[str] | None) → int

src/architecture_model/config/init.py [DORMANT] (25 LOC)

Configuration subpackage for the Architecture Model Standard.

src/architecture_model/config/loader.py [ACTIVE] (558 LOC)

Configuration loader for the Architecture Model Standard.

- load_config(root: Path) → ProjectConfig
- discover_config(root: Path) → tuple[ProjectConfig, DiscoveryReport]
- get_config(root: Path) → ProjectConfig
- write_config(config: ProjectConfig, root: Path | None) → Path

src/architecture_model/config/schema.py [ACTIVE] (325 LOC)

Project configuration schema.

- **class OutputConfig():** 1 methods
- **class ResolvedOutputConfig():** 0 methods
- **class LayerConfig():** 0 methods
- **class SubBlockConfig():** 0 methods
- **class FunctionalBlockConfig():** 0 methods
- **class MetricConfig():** 0 methods
- **class ProjectConfig():** 7 methods
- **class DiscoveryCandidate():** 0 methods
- **class DiscoveryReport():** 3 methods

src/architecture_model/core/init.py [DORMANT] (0 LOC)

src/architecture_model/core/coverage.py [ACTIVE] (268 LOC)

Coverage analysis: compare architecture model against code reality (manifest).

- `coverage_report(model: 'ArchitectureModel', manifest: dict) -> CoverageResult`
- **class CoverageCheck():** 0 methods
- **class CoverageResult():** 1 methods

src/architecture_model/core/decomposer.py [ACTIVE] (659 LOC)

Complexity scoring and system identification for architecture decomposition.

- `compute_complexity(comp: Component, model: ArchitectureModel) -> float`
- `identify_systems(model: ArchitectureModel, manifest: dict) -> list[SystemCandidate]`
- `auto_assign_f_blocks(model: ArchitectureModel, max_cluster_size: int) -> ArchitectureModel`
- `decompose_model(model: ArchitectureModel, manifest: dict, output_dir: str) -> DecompositionResult`
- `test_affinity_decompose(repo_path: Path) -> list[Subsystem]`
- **class SystemCandidate():** 0 methods
- **class Subsystem():** 0 methods
- **class DecompositionResult():** 0 methods

src/architecture_model/core/differ.py [ACTIVE] (294 LOC)

Differ: Compare two architecture model versions and produce a structured diff.

- `diff_models(old_model: ArchitectureModel, new_model: ArchitectureModel) -> ModelDiff`
- **class ChangeType(str, Enum):** 0 methods
- **class EntityChange():** 0 methods
- **class RelationshipChange():** 0 methods
- **class ModelDiff():** 7 methods

src/architecture_model/core/merger.py [ACTIVE] (778 LOC)

Merger: Merge manifest facts into an existing architecture model.

- `merge_manifest(model: ArchitectureModel, manifest_path: str | Path, project_root: str | Path | None) -> ArchitectureModel`
- `enrich_from_manifest(model: ArchitectureModel, manifest: dict) -> EnrichmentResult`
- `compact_for_generation(model: ArchitectureModel) -> ArchitectureModel`
- `compose_enriched_model(project_root: Path) -> ArchitectureModel`
- **class EnrichmentResult():** 4 methods

src/architecture_model/core/parser.py [ACTIVE] (592 LOC)

Parser: Load architecture model YAML, validate against schema, return typed ArchitectureModel.

- `load_model(path: str | Path) -> ArchitectureModel`
- `validate_model_data(data: dict[str, Any]) -> list[str]`
- `dump_model(model: ArchitectureModel) -> dict[str, Any]`
- `save_model(model: ArchitectureModel, path: str | Path) -> None`

src/architecture_model/core/slicer.py [ACTIVE] (496 LOC)

Slicer: Extract subsets of the architecture model.

- `slice_by_fblock(model: ArchitectureModel, f_block: str, include_relationships: bool) -> ArchitectureModel`
- `slice_by_layer(model: ArchitectureModel, layer_id: str) -> ArchitectureModel`
- `slice_by_status(model: ArchitectureModel, status: Status) -> ArchitectureModel`
- `slice_for_artifact(model: ArchitectureModel, artifact_name: str) -> ArchitectureModel`

src/architecture_model/core/types.py [ACTIVE] (665 LOC)

Typed dataclasses for all Architecture Model entity types and relationships.

- **class Status**(str, Enum): 0 methods
- **class RelationType**(str, Enum): 1 methods
- **class ActorType**(str, Enum): 1 methods
- **class InterfaceType**(str, Enum): 1 methods
- **class ConstraintType**(str, Enum): 1 methods
- **class Priority**(str, Enum): 0 methods
- **class Strength**(str, Enum): 0 methods
- **class ComponentKind**(str, Enum): 1 methods
- **class BehaviorPattern**(str, Enum): 1 methods
- **class SymbolKind**(str, Enum): 0 methods
- **class BaseEntity**(): 0 methods
- **class Actor**(BaseEntity): 0 methods
- **class Capability**(BaseEntity): 0 methods
- **class StateTransition**(): 0 methods
- **class Compensation**(): 0 methods
- **class Behavior**(BaseEntity): 0 methods
- **class Endpoint**(BaseEntity): 0 methods
- **class Interface**(BaseEntity): 0 methods

- **class Constraint**(BaseEntity): 0 methods
- **class Layer**(BaseEntity): 0 methods
- **class Symbol**(): 0 methods
- **class DataField**(): 0 methods
- **class Constant**(): 0 methods
- **class FunctionSignature**(): 0 methods
- **class TestContract**(): 0 methods
- **class Component**(BaseEntity): 0 methods
- **class System**(BaseEntity): 0 methods
- **class Relationship**(): 0 methods
- **class ModelMeta**(): 0 methods
- **class Entities**(): 0 methods
- **class ArchitectureModel**(): 5 methods

src/architecture_model/core/validator.py [ACTIVE] (518 LOC)

Validator: Check architecture model invariants.

- `validate_model(model: ArchitectureModel, strict: bool) -> ValidationResult`
- **class Severity**(str, Enum): 0 methods
- **class ValidationIssue**(): 0 methods
- **class ValidationResult**(): 6 methods

src/architecture_model/extract/from_code.py [ACTIVE] (646 LOC)

Extract an ArchitectureModel directly from source code analysis.

- `extract_from_code(project_root: str | Path, config: ProjectConfig | None, manifest: dict | None) -> ArchitectureModel`

src/architecture_model/manifest/init.py [ACTIVE] (58 LOC)

Reality Manifest Generator — modular package.

src/architecture_model/manifest/blocks.py [ACTIVE] (215 LOC)

Functional block definitions and processing.

- `process_block(root: Path, block_id: str, block_def: dict, sub_block_configs: list | None) -> BlockManifest`

src/architecture_model/manifest/body_hints.py [ACTIVE] (270 LOC)

AST-based body hint extraction for function implementations.

- `classify_function(source: str, func_name: str) -> BodyComplexity`
- `extract_body_hint(source: str, func_name: str, class_name: str | None) -> str`
- `extract_file_hints(filepath: Path, include_private: bool) -> list[FunctionSignature]`
- **class BodyComplexity**(Enum): 0 methods

src/architecture_model/manifest/display.py [ACTIVE] (51 LOC)

Terminal summary display for the reality manifest.

- `print_summary(manifest: dict[str, Any]) -> None`

src/architecture_model/manifest/generator.py [ACTIVE] (168 LOC)

Main manifest generation and caching logic.

- `generate_manifest(project_root: Path, config: Optional[Any]) -> Manifest`
- `load_or_generate_manifest(project_root: Path, output_dir: Path | None) -> dict[str, Any]`

src/architecture_model/manifest/interfaces.py [ACTIVE] (89 LOC)

Interface derivation from import analysis between scanned modules.

- `derive_interfaces(modules: list[ModuleInfo], root: Path) -> list[InterfaceEdge]`

src/architecture_model/manifest/metrics.py [ACTIVE] (85 LOC)

Project metrics computation from filesystem scanning.

- `compute_metrics(root: Path, config: Optional[Any]) -> MetricsResult`

src/architecture_model/manifest/scanner.py [ACTIVE] (496 LOC)

AST helpers and file scanning for the reality manifest.

- `scan_file(root: Path, filepath: Path) -> ModuleInfo`

src/architecture_model/manifest/slicers.py [ACTIVE] (579 LOC)

Manifest slicing for artifact context injection.

- `get_manifest_slice(manifest: Manifest | dict[str, Any], artifact_name: str) -> str`

src/architecture_model/manifest/test_analyzer.py [ACTIVE] (501 LOC)

Test Contract Analyzer — extracts behavioral specifications from test files.

- `analyze_test_file(test_file: Path) -> TestAnalysisResult`
- `extract_constants_from_contracts(contracts: list[TestContract]) -> list[Constant]`
- **class TestAnalysisResult():** 0 methods

src/architecture_model/manifest/types.py [ACTIVE] (219 LOC)

Typed dataclasses for manifest generation outputs.

- **class ModuleStatus**(str, Enum): 0 methods
- **class FunctionInfo**(): 0 methods
- **class ClassInfo**(): 0 methods
- **class ImportDetail**(): 0 methods
- **class DecoratedFunction**(): 0 methods
- **class ModuleInfo**(): 1 methods

- **class InterfaceEdge()**: 1 methods
- **class SubFunctionEntry()**: 1 methods
- **class BlockManifest()**: 1 methods
- **class MetricsResult()**: 1 methods
- **class ScanReport()**: 2 methods
- **class Manifest()**: 1 methods

src/architecture_model/profiles/init.py [DORMANT] (1 LOC)

Domain profiles for cross-domain architecture modeling.

src/architecture_model/profiles/builtins/init.py [DORMANT] (0 LOC)

src/architecture_model/profiles/schema.py [ACTIVE] (85 LOC)

Domain profile schema and loading logic.

- `load_profile(name_or_path: str) -> DomainProfile`
- **class EnumExtension()**: 0 methods
- **class EntityExtension()**: 0 methods
- **class ConditionalRule()**: 0 methods
- **class DomainProfile()**: 2 methods

src/architecture_model/spec/init.py [DORMANT] (0 LOC)

src/architecture_model/utis/init.py [DORMANT] (1 LOC)

Shared utilities for architecture-model-standard.

src/architecture_model/utis/discovery.py [ACTIVE] (72 LOC)

Shared file discovery and exclusion logic.

- `is_excluded_dir(path: Path) -> bool`
- `collect_py_files(directory: Path, recursive: bool, exclude_init: bool) -> list[Path]`
- `discover_source_files(project_root: Path) -> list[Path]`
- `discover_test_files(project_root: Path) -> list[Path]`